

Big Data Analytics

Big data analytics is the process of examining large and varied data sets --i.e., big data -- to uncover **hidden patterns, unknown correlations, market trends, customer preferences** and other useful information that can help organizations make more-informed business decisions.

Analytical finding can lead to more effective **marketing, better customer service & satisfaction, newer products & services, improved operational efficiency, reduced expenditure** etc.. (ex: Amazon recommendations)

Recommendation systems in ecommerce

- Make your marketing unique to every customer
- Learn about your audience and tailor your messaging to create personalized experiences for all your customers.
- Suppose a customer buys new running shoes every three months. They always buy the same brand. If this customer comes back to the site and searches for running shoes, it's a good guess that they will buy their favorite brand again—but maybe they would like new features, a different color, or a shoe with new product endorsement.
- A personalized product recommendation for this customer keeps old choices available, shows the possibility of a "new and improved" customer experience at a different price point first.

Recommendation systems in ecommerce

• Recommendation engines usually look at data such as:

- A customer's search queries
- Their purchase history
- What's currently in their shopping cart
- Their social behavior (likes, shares, etc.)
- Their geographic location
- The customer's audience segment (demographics)

Data → Raw facts collected from multiple sources (databases, logs, social media, etc.).

Information → Processed and structured form of data, presented in reports, dashboards, or charts.

Insight → Analysis of information to identify patterns, trends, comparisons, and conclusions.

Actionable Insight →

The most valuable form of insight.

Provides clear, practical direction.

Encourages immediate decisions and actions.

Delivers maximum business value from data analytics.

.Actionable insights are insights that deliver immense value, provide a clear direction, and drive instant actions towards achieving business objectives.

. These insights are the most valuable and provide **maximum returns on data analytics investments.**

.Actionable insights are distilled from raw data in the following way.

.Data: Raw data is collected from multiple sources such as information databases, spreadsheets, plans, social media comments, call logs, and so on. This data can be structured (forms and tables) or unstructured (log files, comments) and qualitative (observations) or quantitative (numbers or units).

. Information: Raw data, when processed, categorized, and organized, is converted into information. This information can be presented in the form of reports, dashboards, charts, and other visualizations.

.Insight: Information is analyzed to identify patterns, summarize findings, and draw conclusions in the form of insights. Insights help to understand business events, answer questions, compare details, spot similarities and differences, group similar items, and track trends.

.Actionable Insight: Among the various insights that emerge from data, actionable insights are crucial. These insights encourage informed actions and decisions based on data-driven recommendations. The control to take action or not rests with the user.

Benefits of Actionable Insights

.Enhance Decision Making: Actionable insights give business users the much-needed support to enhance their decision-making. When insights are aligned with an organization's business objectives and KPIs, they become more actionable and influence better decision-making.

.Actionable insights provide decision intelligence on what matters most to decision-makers based on their role, work, or goal.

Benefits of Actionable Insights

.Understand Context Better: Actionable insights go beyond answers to provide context for changing business metrics.

.For example, the monthly report shows a surge in sales by 20% over the previous month. It may look good now, but it is missing a context.

. The surge was due to an increase in the price of goods, while the number of transactions actually went down.

. Such a surge might not help in the long run. This actionable insight alerts the management that the price rise is driving away customers, and they can take measures to win back customers.

. Actionable insights explain not only what happened but also why it happened and how to proceed ahead.

Gain Clarity

- While data processing and insight extraction can be handled using automated analytics, machine learning, and AI algorithms, the true value comes from **presenting those insights clearly**.
- Actionable insights become more meaningful and useful when displayed in **interactive charts and visualizations**, allowing users to explore the data in ways that best suit their needs.

Get Specific Recommendations

Actionable insights filter out the **noise from lengthy reports, multiple dashboards**, and overwhelming data to provide **clear and focused guidance**.

For example,

- An actionable insight highlighting **growing demand from a new customer segment**,
- Presented as an automated business headline,
- Enables customer success managers to **develop targeted offerings and strengthen customer relationships**.

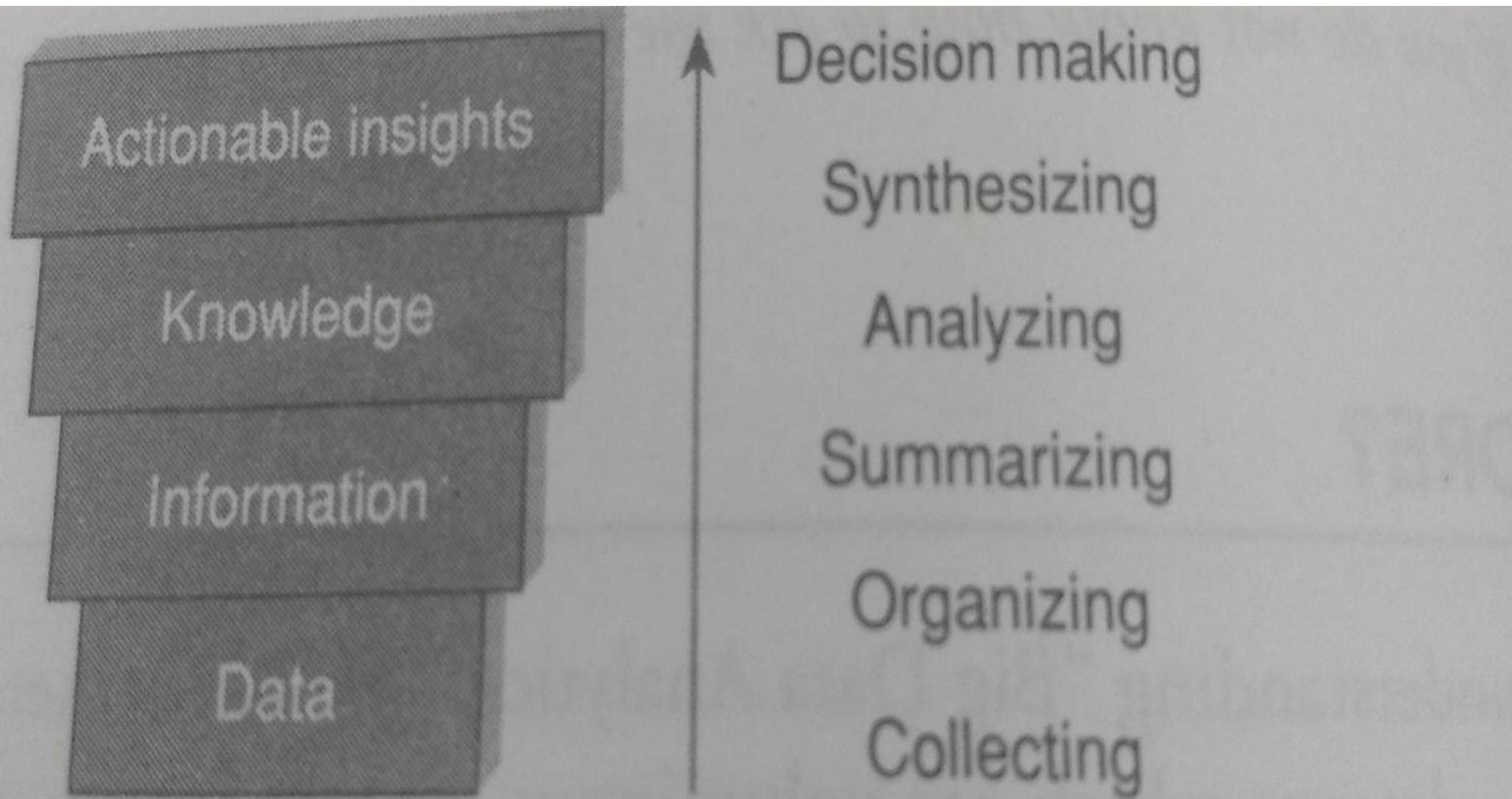
Save Significant Time and Efforts:

.When users get access to relevant, focused, and personalized insights, they can understand business situations accurately and make quick decisions.

. With actionable insights, users no longer have to go through volumes of information and unnecessary vanity insights.

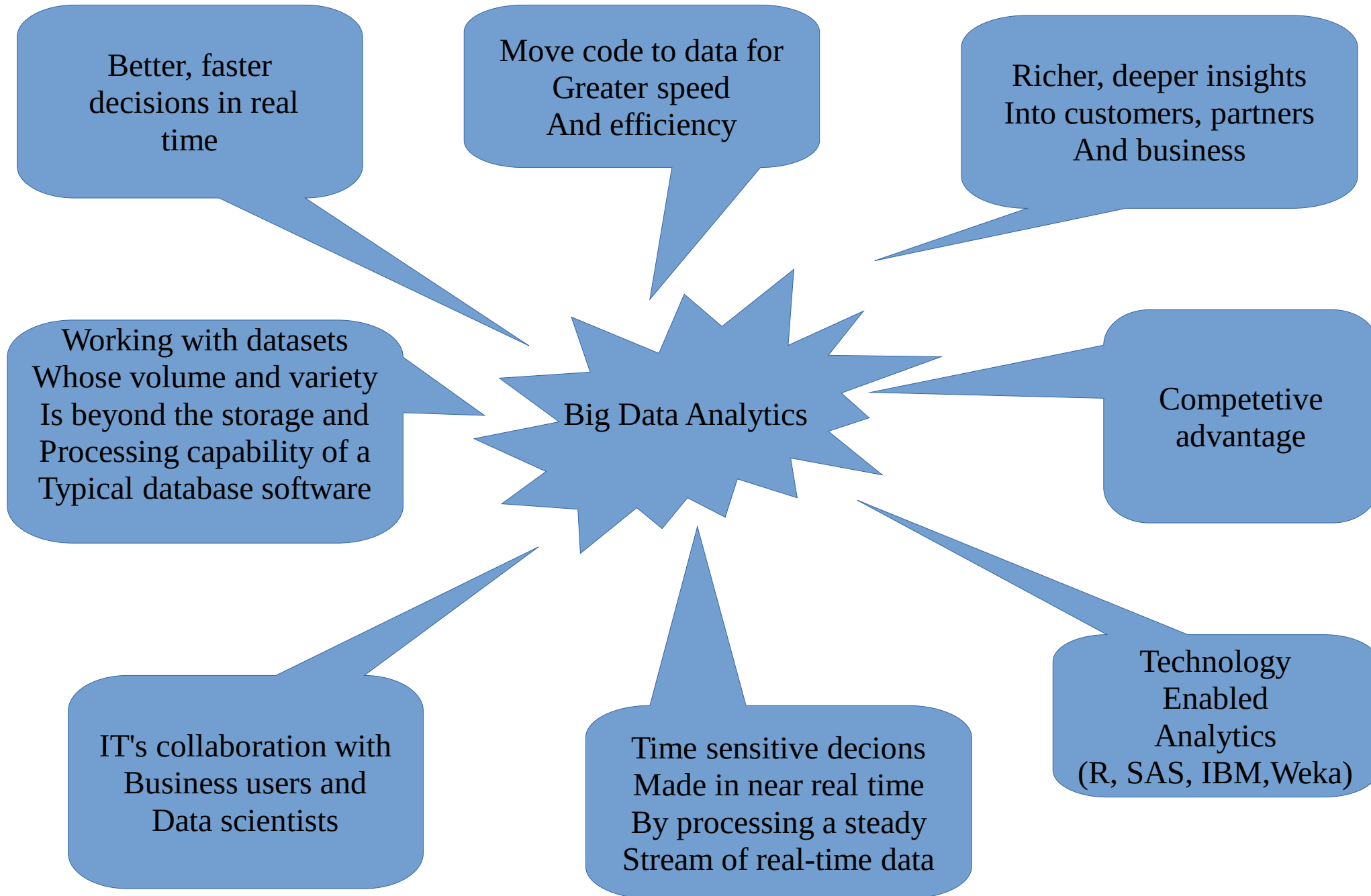
. With modern data analytics platforms users can access data easily and get insights in an automated way without investing time and effort in learning complex techniques or depending on experts.

Where Do We Begin ?

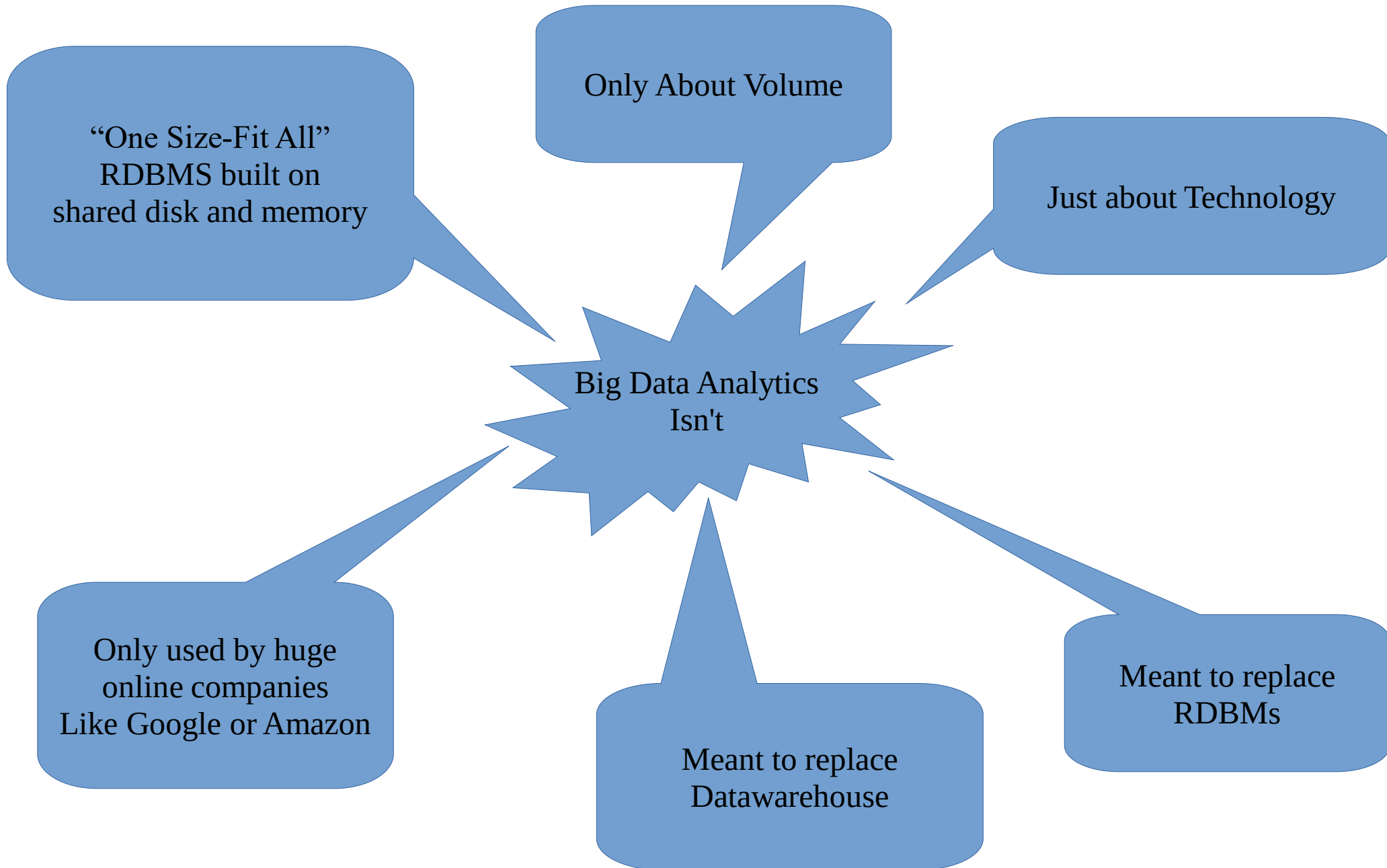


Transformation of data to yield actionable insights

What is Big Data Analytics?



What is Big Data Analytics isn't?



Why sudden hype around BDA ?

- Data growing at 40% compound rate, reaching nearly 50 ZB by 2020
- Cost per GB of storage has hugely dropped
- There is a shortage of user friendly analytical tools available in the market